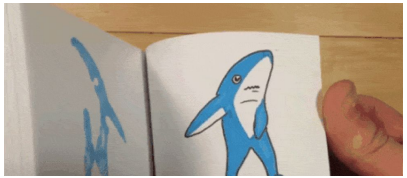
Technical Stuff & Tool Box

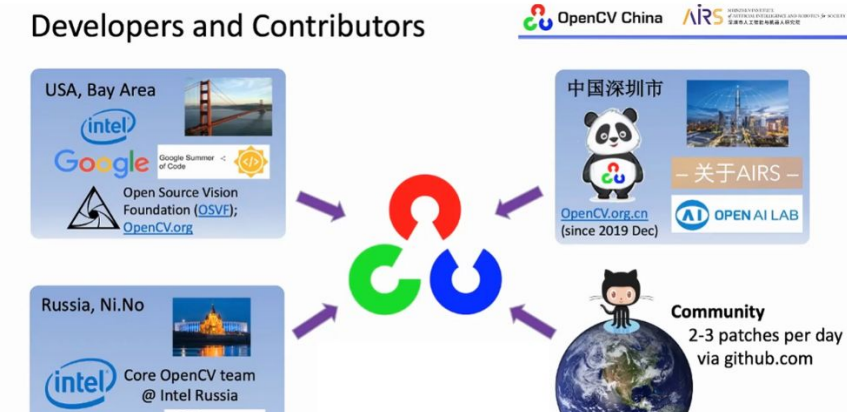
What Are We Doing Today? High level



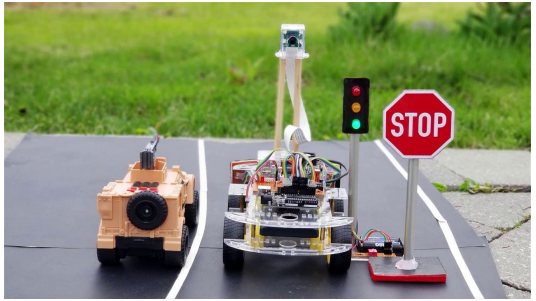
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

OpenCV: What And Why

- Mature (**2000 – present**)
- Free for research and commercial use (**BSD license -> Apache 2 license (5.x)**)
- Implemented in **C/C++** with automatically generated bindings for **Python, Java, Javascript**
- Big, but modular: **>1M** lines of code; **70+** modules (**opencv** + **opencv_contrib**)
- Downloaded **21M** times from SourceForge; **10K** times/week from http://github/opencv



Things Made With OpenCV



Stanley, an autonomous car developed by the Stanford University Racing Team, won the 2005 DARPA Grand Challenge after successfully traversing a 132-mile course. Photo courtesy of the Smithsonian Air & Space Museum.

Other tools

Numpy

Mediapipe

Yolo models

Self Learning: OpenCV and Beyond

CV Basics

Computer Vision: Crash Course Computer Science #35

Decent 11 minute overview of how the deep learning algos we will automagically run over our images work.

Applied Basics

https://opencv.org/university/free-opencv-course/?utm_source=opcv&utm_medium=menu&utm_campaign=obc <- opencv intro

From the makers of opencv, a series of short how to videos

Mid level data processing

https://numpy.org/doc/stable/user/absolute_beginners.html

With image processing we use numpy instead of pandas, why?

The prime use of dataframes is a matrix like interface for mixed type data, images have one data type, so we don’t need to sacrifice the speed of numpy

Inspo

https://learnopencv.com/ai-fitness-trainer-using-mediapipe/

https://learnopencv.com/building-a-body-posture-analysis-system-using-mediapipe/